Set Counting Through Residues, Recurrences, and Finite Fourier Analysis

N-20220808

Residue classes turn arithmetic constraints on large sets of integers into small finite combinatorial problems. Sometimes this gives an extremal construction; sometimes it gives an exact count. A particularly effective method is to encode every subset by a monomial, then use roots of unity to select those subsets whose sums satisfy a prescribed congruence. The resulting formula can also be read as a diagonalization of a recurrence on residue classes.[1]

§1.1 An extremal construction becomes a graph on residues

Consider integers $1, \dots, 11m$. Choose a set A so that no two elements of A have residues differing by 1 or -1 modulo 11. Elements with the same residue are allowed.

The eleven residue classes can be arranged around a cycle:

$$0-1-2-\cdots-10-0.$$

If two adjacent vertices were occupied by residues of elements of A, the condition would fail. Thus the occupied residue classes form an independent set in the cycle graph C_{11}.

An independent set of k vertices in an odd cycle of length 11 leaves at least k unoccupied neighbors, so $2k \le 11$ and $k \le 5$. Equality is achieved, for instance, by the residues $\{0, 2, 4, 6, 8\}$. Each residue appears exactly m times among $1, \dots, 11m$, hence

$$\boxed{\max |A| = 5m.}$$

The extremal argument has two distinct pieces: a quotient of the integers identifies the finite constraint graph, and a sharp construction occupies every element in the five selected residue classes. The graph exposes the combinatorial obstruction without requiring a search over $11m$ individual integers.

§1.2 Encode all subsets at once

For $n \ge 1$, let $N_r(n)$ denote the number of subsets $S \subseteq \{1, \dots, n\}$ with

$$\sum_{k \in S} k \equiv r \pmod 3, \qquad r \in \{0, 1, 2\}.$$

The empty subset is included, with sum zero.

Introduce the polynomial

$$P_n(z) = \prod_{k=1}^{n} (1 + z^k).$$

[1] For the root-of-unity filter and its discrete Fourier interpretation see the University of Illinois Chicago notes on discrete Fourier transforms: https://homepages.math.uic.edu/~jan/mcs472/mcs472notes/lec09.html; for generating-function methods see Stanley, *Enumerative Combinatorics*, Volume 1.

In each factor, choosing 1 means omitting k, while choosing z^k means including k. Therefore

$$P_n(z) = \sum_{S\subseteq\{1,\dots,n\}} z^{\sum_{k\in S} k}.$$

The coefficient of z^j counts the subsets whose sum is exactly j. Evaluating at $z = 1$ gives the familiar count $P_n(1) = 2^n$.

Our problem requires grouping exponents modulo 3. It would be wasteful to calculate all coefficients and add one third of them. We need an operation that keeps precisely one residue class of exponents.

§1.3 The roots of unity are perfect residue detectors

Let $\omega = e^{2\pi i/3}$, so

$$1 + \omega + \omega^2 = 0, \qquad \omega^3 = 1.$$

The geometric-sum identity gives

$$\frac{1}{3}(1 + \omega^j + \omega^{2j}) = \begin{cases} 1, & j \equiv 0 \pmod 3, \\ 0, & j \not\equiv 0 \pmod 3. \end{cases}$$

This formula is the indicator function of a congruence condition, expressed using complex numbers.

Apply it term by term to P_n:

$$\begin{aligned} N_0(n) &= \sum_{S\subseteq[n]} \frac{1}{3}(1 + \omega^{\sum_{k\in S} k} + \omega^{2\sum_{k\in S} k}) \\ &= \frac{1}{3}(P_n(1) + P_n(\omega) + P_n(\omega^2)). \end{aligned}$$

Thus

$$\boxed{N_0(n) = \frac{2^n + P_n(\omega) + P_n(\omega^2)}{3}.}$$

The large combinatorial count has become a small number of special polynomial evaluations.

§1.4 An exact formula for all n

Group consecutive integers into blocks of three. Since $\omega^3 = 1$,

$$\begin{aligned} (1 + \omega^{3j+1})(1 + \omega^{3j+2})(1 + \omega^{3j+3}) &= (1 + \omega)(1 + \omega^2) \cdot 2 \\ &= 2. \end{aligned}$$

The same is true at ω^2. Consequently, for $n = 3m$,

$$P_{3m}(\omega) = P_{3m}(\omega^2) = 2^m, \qquad \boxed{N_0(3m) = \frac{2^{3m} + 2^{m+1}}{3}.}$$

For $n = 3m + 1$, the extra factor has residue 1, and

$$\begin{aligned} P_n(\omega) + P_n(\omega^2) &= 2^m((1 + \omega) + (1 + \omega^2)) \\ &= 2^m. \end{aligned}$$

Therefore

$$\boxed{N_0(3m+1) = \frac{2^{3m+1} + 2^m}{3}.}$$

For $n = 3m + 2$, the two extra factors $(1 + \omega)(1 + \omega^2)$ have product 1, so

$$\boxed{N_0(3m+2) = \frac{2^{3m+2} + 2^{m+1}}{3}.}$$

These three formulas cover every nonnegative n, taking $N_0(0) = 1$.

For a quick check, the subsets of $\{1, 2, 3\}$ whose sums are multiples of three are

$$\varnothing, \quad \{3\}, \quad \{1, 2\}, \quad \{1, 2, 3\}.$$

There are four, in agreement with $(2^3 + 2^2)/3 = 4$.

§1.5 Induction Meets Fourier Analysis

Every subset of $\{1, \ldots, n\}$ either excludes n or includes it. Thus

$$\boxed{N_r(n) = N_r(n-1) + N_{r-n \bmod 3}(n-1),}$$

where residue indices are read modulo three. The initial vector is

$$(N_0(0), N_1(0), N_2(0)) = (1, 0, 0).$$

This recurrence could be used for a direct induction proof of the formulas above.

The deeper structure becomes visible in vector notation. Write

$$v_n = \begin{pmatrix} N_0(n) \\ N_1(n) \\ N_2(n) \end{pmatrix}, \qquad (Sv)_r = v_{r-1 \bmod 3}.$$

Then

$$\boxed{v_n = (I + S^n)v_{n-1}.}$$

The matrix S cyclically permutes residue classes. Its eigenvalues are $1, \omega, \omega^2$. Passing to its eigenbasis, the discrete Fourier transform, diagonalizes every factor $I + S^n$. The eigenvalue products are exactly

$$P_n(1), \quad P_n(\omega), \quad P_n(\omega^2).$$

The roots-of-unity formula is therefore the closed-form solution of the induction recurrence in a better coordinate system.

§1.6 Why subset sums become nearly uniform

There are 2^n subsets in total, so a perfectly uniform distribution of their sums modulo three would place $2^n/3$ subsets in each class. The formulas show that the discrepancy for residue zero is only

$$N_0(n) - \frac{2^n}{3} = \begin{cases} 2^{m+1}/3, & n = 3m, \\ 2^m/3, & n = 3m+1, \\ 2^{m+1}/3, & n = 3m+2. \end{cases}$$

It grows like $2^{n/3}$, while the total number of subsets grows like 2^n. The relative error therefore decays exponentially.

In Fourier language, the constant mode $P_n(1) = 2^n$ determines the uniform component. The nontrivial modes $P_n(\omega)$ and $P_n(\omega^2)$ measure imbalance between residue classes, and their relative size shrinks rapidly.

This example explains the value of the finite Fourier transform beyond a clever counting trick. A modular condition makes the problem finite, a generating function packages all subsets simultaneously, and the Fourier basis separates the uniform count from the oscillatory corrections. Induction then reappears as the same operation written in the original residue coordinates.